

# SILICONA

## TECHNICAL EVIDENCE DOSSIER

Evidence state, engineering record, certification ladder, benchmark corpus, and infrastructure requirements

Prepared for  
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Document date  
1 October 2026

Classification  
CONFIDENTIAL - FOUNDER / TECHNICAL REVIEW

**Purpose: provide a clinical record of what SILICONA reports as implemented, tool-verified, artifact-bound, and not yet independently reproduced.**

AI proposes. EDA calculates. Verification decides. Engineer approves.

### Source basis

This document is derived from the supplied SILICONA Independent Evaluation Report dated 30 September 2026 and the accompanying corrective review. Figures and statuses are reproduced as supplied. Claims are separated by evidence state, and unsupported superlatives or inferred competitor capabilities are excluded.

# 1. Scope and evidence discipline

SILICONA is presented here as an AI-native semiconductor engineering control plane whose governing doctrine is evidence before narrative. The supplied record links AI-generated engineering proposals to deterministic EDA execution, verification gates, artifacts, hashes, benchmark results, and recorded failures.

This dossier does not establish that SILICONA is superior to another company or tool. It records the supplied evidence corpus and identifies the boundaries of what the record supports.

The strongest current evidence concerns an open-PDK physical implementation workflow, a certification state machine, a longitudinal artifact ledger, and measured design-space search results. The principal unresolved scaling issue is compute capacity.

# 2. Evidence hierarchy

Every material statement in this dossier is classified against the following hierarchy. E4 is a future target, not a current achievement.

| Level                         | Definition                                                                          | SILICONA example                                                     |
|-------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| E0 - Claimed                  | Statement without tool-verified backing                                             | General claim about handling complex SoCs                            |
| E1 - Implemented              | Code, registry, or workflow exists without a qualifying measured workload           | Provider registry exists; cross-provider measurement remains pending |
| E2 - Tool-verified            | A deterministic tool produces a recorded verdict on an artifact                     | SymbiYosys formal PASS or OpenSTA timing result                      |
| E3 - Artifact-bound           | The verdict is bound to revision, toolchain, seed or run context, and artifact hash | DRC 0 associated with a named revision and KLayout run               |
| E4 - Independently reproduced | A third party replays the published manifest and obtains the same verdict           | Not yet achieved; Phase C target                                     |

# 3. Core doctrine

The supplied record describes the doctrine as an enforcement model rather than a branding statement.

| Doctrine element     | Reported status | Evidence basis                                                                                 | Current level |
|----------------------|-----------------|------------------------------------------------------------------------------------------------|---------------|
| AI proposes          | Implemented     | GLM-4-Plus primary reasoning agent and multi-provider registry                                 | E1            |
| EDA calculates       | Implemented     | Yosys 0.68, SymbiYosys + Z3, iverilog 14.0, OpenSTA 2.6.0, OpenROAD v2.0-17598, KLayout 0.30.0 | E2            |
| Verification decides | Implemented     | Certification transitions require deterministic tool verdicts                                  | E2            |
| Engineer approves    | Implemented     | Policy-controlled approval gates with bounded repair cycles                                    | E1/E2         |

# 4. Explicit 14-stage engineering pipeline

The original evaluation listed more than 14 named activities while referring to a 14-stage pipeline. The corrected representation below defines the grouping explicitly and removes the ambiguity.

| Stage                  | Activities contained                    |
|------------------------|-----------------------------------------|
| 1. Specification       | Requirement capture, interface contract |
| 2. Architecture        | Decomposition, block partitioning       |
| 3. RTL Generation      | AI-proposed Verilog, lint               |
| 4. Simulation          | iverilog testbench execution            |
| 5. Formal Verification | SymbiYosys + Z3, BMC/prove              |

| Stage                      | Activities contained                              |
|----------------------------|---------------------------------------------------|
| 6. Synthesis               | Yosys mapping to SKY130 liberty                   |
| 7. Equivalence             | RTL versus netlist check                          |
| 8. Floorplan               | OpenROAD floorplan initialization                 |
| 9. Placement               | OpenROAD global and detailed placement            |
| 10. Clock Tree Synthesis   | OpenROAD CTS                                      |
| 11. Routing                | OpenROAD global and detailed routing              |
| 12. Timing Analysis        | OpenSTA on mapped/routed netlist                  |
| 13. Physical Verification  | KLayout DRC + LVS on GDS                          |
| 14. Report & Certification | Evidence packaging and certification state update |

## 5. Open-PDK physical implementation record

The supplied evidence reports a complete qualifying run through the defined 14-stage grouping for the reference FIFO workload. The appropriate terminology is demonstrated SKY130 open-PDK physical closure and physical implementation and verification. The record does not support the broader commercial term physical signoff.

| Metric                 | Reported result                          | Evidence level |
|------------------------|------------------------------------------|----------------|
| Pipeline stages passed | 14 / 14                                  | E2/E3          |
| DRC violations         | 0                                        | E2/E3          |
| LVS verdict            | MATCH, dual checks reported              | E2/E3          |
| Devices                | 20,962                                   | E3             |
| Nets                   | 11,544                                   | E3             |
| Pins                   | 74                                       | E3             |
| Timing slack           | +5.74 ns pre-route / +3.48 ns post-route | E2/E3          |
| Mapped cells           | 950                                      | E3             |
| Cell area              | 18,672.9 $\mu\text{m}^2$                 | E3             |
| Fmax                   | 240 MHz                                  | E2/E3          |
| GDS size               | 2,361,570 bytes                          | E3             |
| Reproducibility corpus | 24 revisions                             | E3             |

**Known boundary: the supplied record states that post-route extraction is estimated rather than SPEF-extracted, and hold fixing and IR-drop analysis are not yet part of the flow. These gaps are material to any claim of commercial signoff readiness.**

## 6. Certification ladder

Certification state is separated from project existence. The supplied record reports five of ten certification levels as currently certified.

| Level | Definition                       | Status    | Supplied evidence                            |
|-------|----------------------------------|-----------|----------------------------------------------|
| L1    | Combinational block              | CERTIFIED | Priority Encoder 8-3, lint + simulation PASS |
| L2    | Stateful block with formal proof | CERTIFIED | Sync FIFO 16x32, SymbiYosys + Z3 formal PASS |
| L3    | Protocol IP, UART TX 8N1         | CERTIFIED | Full defined pipeline PASS                   |

| Level | Definition                            | Status    | Supplied evidence                                                  |
|-------|---------------------------------------|-----------|--------------------------------------------------------------------|
| L4    | Optimized compute block, ALU          | TEMPLATE  | Project exists, qualifying run pending                             |
| L5    | Subsystem integration                 | CERTIFIED | UART + FIFO + registers, full pipeline PASS                        |
| L6    | Processor, RV32I                      | TEMPLATE  | Project exists, qualifying run pending                             |
| L7    | Accelerator discovery                 | CERTIFIED | Grid SearchRun winner passes full pipeline                         |
| L8    | Complete SoC                          | TEMPLATE  | False-claim audit blocked certification after 3 equivalence errors |
| L9    | Complete physical implementation      | CERTIFIED | Rev-22/24, DRC 0, dual LVS MATCH, GDS streamout                    |
| L10   | Architectures beating human baselines | MEASURED  | Honest negatives included; one RL stream-fifo result was 0.00%     |

## 7. False-certification control event

The L8 incident is important as governance evidence. Three equivalence errors blocked a certification state rather than being hidden or overridden. The event supports the existence of a deterministic gate between an engineering claim and a certification state.

This dossier does not treat the incident as proof of broad system reliability. It is evidence of one observed control behavior under the supplied run.

## 8. AI model evaluation ledger

The current measured corpus is concentrated on a single model provider. This is both a measurement result and a dependency.

| Metric                 | Reported result                          | Evidence level |
|------------------------|------------------------------------------|----------------|
| Total AI evaluations   | 177                                      | E3             |
| Provider concentration | glm/glm-4-plus, 100% of current corpus   | E3             |
| Successful evaluations | 131                                      | E3             |
| Partial evaluations    | 1                                        | E3             |
| Failed evaluations     | 51, HTTP 429 rate limits                 | E3             |
| Total tokens           | ~147K                                    | E3             |
| Mean response time     | 2.3 seconds                              | E3             |
| Failures retained      | Yes, all 51 rate-limit failures recorded | E3             |

## 9. Design-space search evidence

The search corpus contains both positive and negative outcomes. The results below are reported as supplied and should be read with their sample sizes.

| Family      | Mode  | n | Reported result                          | Evidence level |
|-------------|-------|---|------------------------------------------|----------------|
| matrix-mac  | multi | 5 | +98.21%, reported $\pm 0.00$ across runs | E3             |
| fir-filter  | multi | 5 | Average +79.65%, range 74.15% to 83.32%  | E3             |
| stream-fifo | multi | 6 | +34.30%                                  | E3             |
| stream-fifo | rl    | 1 | 0.00%, retained as an honest negative    | E3             |
| matrix-mac  | grid  | 2 | FAILED, published as failed              | E3             |

## 10. Engineering memory and provenance

The evidence record is longitudinal rather than a collection of selected screenshots. The supplied metrics describe a growing corpus of artifacts, stage runs, failure classifications, and revision history.

| Metric                         | Reported value                       | Evidence level |
|--------------------------------|--------------------------------------|----------------|
| Hash-bound artifacts           | 1,138                                | E3             |
| Ledger content                 | 91.17 MB                             | E3             |
| Database density per revision  | ~0.76 MB                             | E3             |
| Working tree size per revision | 35 to 40 MB                          | E3             |
| Search scratch per candidate   | 0.32 MB, 288 CAS-deduped directories | E3             |
| Stage runs recorded            | 563                                  | E3             |
| Stage pass rate                | 81.0%                                | E3             |
| Flow failures with root causes | 104                                  | E3             |

## 11. Infrastructure readiness and measured constraints

The current development environment is reported at approximately 4 GB RAM, 10 GB disk, and 2 CPU cores. The supplied record reports OOM kills, Turbopack panics, and reload friction. These are infrastructure constraints documented against workload requirements.

The production BOM translates the workload into target resources. It does not establish that those resources have been provisioned.

| Resource | Current environment       | Production target in supplied BOM                           |
|----------|---------------------------|-------------------------------------------------------------|
| RAM      | ~4 GB                     | 256 GB+ solver fleet; 512 GB to 1 TB physical design        |
| Disk     | ~10 GB, reported 94% used | 4 to 8 TB NVMe scratch; 25 TB+ object storage in Y1         |
| CPU      | 2 cores                   | 64+ cores formal; 32+ cores synthesis                       |
| GPU      | 0                         | 1x H200 dense inference; 8x H200 full MoE; 1x 8-GPU RL node |
| Network  | Sandbox                   | 25 to 100 Gbps physical-design I/O                          |

## 12. Production infrastructure BOM

The supplied year-one infrastructure estimate totals approximately \$150,000. These figures are an engineering planning estimate, not a verified procurement quote.

| Pool                             | Profile                                                         | Year-1 estimate                  |
|----------------------------------|-----------------------------------------------------------------|----------------------------------|
| A. Control plane and engineering | API 8 vCPU / 32 GB, HA orchestrator, DB 16c / 64 GB / 1 TB NVMe | ~\$12K                           |
| B. CPU high-memory solver fleet  | 2x 32 cores, >=3.5 GHz, 256 GB, NVMe + burst                    | ~\$44K                           |
| C. Physical design workers       | 64 cores, 512 GB to 1 TB, 4 to 8 TB NVMe, 25 to 100 Gbps        | ~\$8K burst, ~\$40K steady-state |
| D. AI/GPU workers                | 1x H200 dense, 8x H200 full MoE, 1x 8-GPU RL node               | ~\$35K                           |
| E. Object storage                | 25 TB Y1 to 100 TB Y3, CAS-backed and tiered                    | ~\$8K                            |
| TOTAL                            | Year-one planning envelope                                      | ~\$150K                          |

## 13. Phase 0 controls before external compute

The supplied review identifies six code-level prerequisites. These controls belong ahead of distributed execution so infrastructure expansion does not amplify existing resource and storage failure modes.

| Priority | Control                                 | Reason stated in source                                       |
|----------|-----------------------------------------|---------------------------------------------------------------|
| 1        | Global concurrency limiter + queue      | Reported unbounded behavior; search.ts:757 bypasses dedup map |
| 2        | Disk-space guard pre-run                | Prevent ENOSPC cascades                                       |
| 3        | Blob store for artifacts above 900 KB   | ARTIFACT_MAX truncates ledger content at pipeline.ts:114      |
| 4        | Timeout retuning by hardware class      | 720s OpenROAD timeout sized for 2-thread conditions           |
| 5        | Postgres at first multi-node deployment | Shared state requirement at scale                             |
| 6        | Automated scratch tiering               | Control ephemeral workspace growth                            |

## 14. Known risks and current mitigations

| Risk                                | Severity in supplied assessment | Current mitigation or next action                                                        |
|-------------------------------------|---------------------------------|------------------------------------------------------------------------------------------|
| Single-provider dependency          | HIGH                            | Phase B cross-provider measurement; provider registry                                    |
| Sandbox compute ceiling             | HIGH                            | Compute Federation and infrastructure partnerships                                       |
| Unregistered legal status           | MEDIUM                          | Establish suitable corporate structure and assign IP                                     |
| Founder concentration               | MEDIUM                          | Automate, document, hire verification, distributed systems, security                     |
| False certification risk            | LOW                             | Certification state machine and L8 audit                                                 |
| Cloud credit expiration             | MEDIUM                          | Multi-provider execution and portable workloads                                          |
| Open-source EDA limitations         | LOW                             | Open stack as substrate, commercial EDA integration for advanced nodes                   |
| Export-control / sanctions exposure | MEDIUM                          | Model-agnostic routing and sovereign compute options, subject to applicable requirements |

## 15. What is not yet proven

Independent third-party reproduction, E4, has not been achieved.

The current AI evaluation corpus is 100% concentrated on GLM-4-Plus. Cross-provider performance has not yet been measured in the supplied corpus.

Commercial signoff-grade methodology is not established by the current record. SPEF-based extraction, hold fixing, IR-drop analysis, and other signoff requirements remain outside the documented flow.

The supplied record does not establish silicon tapeout, manufactured silicon yield, or measured post-silicon behavior.

The supplied record does not establish broad enterprise customer traction or recurring commercial revenue.

The supplied record does not establish internal capabilities, performance, or shipment status of competing companies beyond what their cited public materials contain.

## 16. Evidence Graph

The proposed long-term technical asset is the Evidence Graph. It links an AI proposal to the design revision, tool execution, failure, repair, verification verdict, physical implementation, artifact, hash, benchmark result, and historical comparison.

The graph becomes more useful as repeated workloads create longitudinal evidence. The commercial consequence described in the review is trust asymmetry: a buyer receives a traceable engineering record rather than a model-generated performance narrative.

AI proposal -> design revision -> deterministic tool run -> failure or pass -> repair -> verification -> physical implementation -> artifact -> hash -> benchmark -> historical comparison

## 17. Recommended evidence publication package

For external technical review, publish the Benchmark Protocol, the artifact manifest, toolchain versions, run identifiers, relevant hashes, failure classifications, and a reproducibility manifest. Mark every external result with its evidence level.

For external infrastructure discussions, pair this dossier with the production BOM and a concise demonstration. The infrastructure request should be framed as a measured workload requirement, not as a claim of entitlement to capacity.

## 18. Technical conclusion

The supplied evidence supports a working engineering workflow with deterministic EDA integration, a certification state machine, a substantial artifact corpus, measured design-space search results, and demonstrated SKY130 open-PDK physical implementation and verification.

The evidence also exposes the current boundaries: limited compute, single-provider AI measurement, incomplete commercial signoff methodology, and no E4 independent reproduction yet.

The appropriate next step is not broader narrative. It is broader measurement, stronger infrastructure controls, independent reproduction, cross-provider evaluation, and progression from artifact-bound evidence toward externally reproducible evidence.